

Module 2

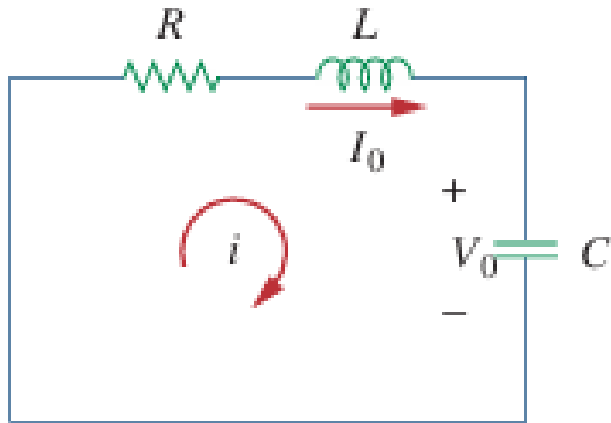
Transient Analysis of First and Second-Order Circuits

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Source Free RLC series circuit

The circuit is being excited by the energy initially stored in the capacitor and inductor. The energy is represented by the initial capacitor voltage V_0 and initial inductor current I_0 .



At $t = 0$,

$$v(0) = \frac{1}{C} \int_{-\infty}^0 i \, dt = V_0 \quad \text{--- 1}$$

$$i(0) = I_0 \quad \text{--- 2}$$

Applying KVL around the loop,

$$Ri + L \frac{di}{dt} + \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = 0$$

To eliminate the integral, we differentiate with respect to t and rearrange terms.

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0 \quad \text{--- 3}$$

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To solve this second-order differential equation we require two initial conditions, the initial value of i and its first derivative or initial values of some i and v .

Initial value of i is $i(0) = I_0$ --- 4

We get the initial value of the derivative of i from $Ri(0) + L\frac{di(0)}{dt} + V_0 = 0$

$$\frac{di(0)}{dt} = -\frac{1}{L}(RI_0 + V_0) \quad \text{--- 5}$$

We can now solve eqn.3. Let us assume $i = Ae^{st}$ --- 6

where A and s are constants to be determined. Substituting Eqn.6 into Eqn.3 and carry out the necessary differentiations.

$$As^2e^{st} + \frac{AR}{L}se^{st} + \frac{A}{LC}e^{st} = 0$$

$$Ae^{st}\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right) = 0$$

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$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 \quad \text{--- 7}$$

This quadratic equation is known as the characteristic equation of the differential Eqn.3. The two roots are:

$$s_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$
$$s_2 = -\frac{R}{2L} - \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \quad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} \quad \text{--- 8}$$

$$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

The roots s_1 and s_2 are called natural frequencies, measured in nepers per second (Np/s). They are associated with the natural response of the circuit.

ω_0 is known as the resonant frequency or strictly as the undamped natural frequency, expressed in radians per second (rad/s).

α is the neper frequency or the damping factor, expressed in nepers per second.

Eqn.7 can now be expressed as, $s^2 + 2\alpha s + \omega_0^2 = 0$

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The two values of s in Eqn.8 indicate that there are two possible solutions for i , each of which is of the form,

$$i_1 = A_1 e^{s_1 t}, \quad i_2 = A_2 e^{s_2 t}$$

Eqn.3 is a linear equation, so, any linear combination of the two distinct solutions i_1 and i_2 is also a solution of Eqn.3. A complete or total solution of Eqn.3 would therefore require a linear combination of i_1 and i_2 . Thus, the natural response of the series RLC circuit is;

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{--- 9}$$

where the constants and are determined from the initial values $i(0)$, $di(0)/dt$ from Eqn.2 and eqn.5 respectively.

1. If $\alpha > \omega_0$, we have the *overdamped* case. Roots of the characteristic eqn are unequal and real.
2. If $\alpha = \omega_0$ we have the *critically damped* case. Roots of the characteristic eqn are equal and real
3. If $\alpha < \omega_0$ we have the *underdamped* case. Roots of the characteristic eqn are complex.

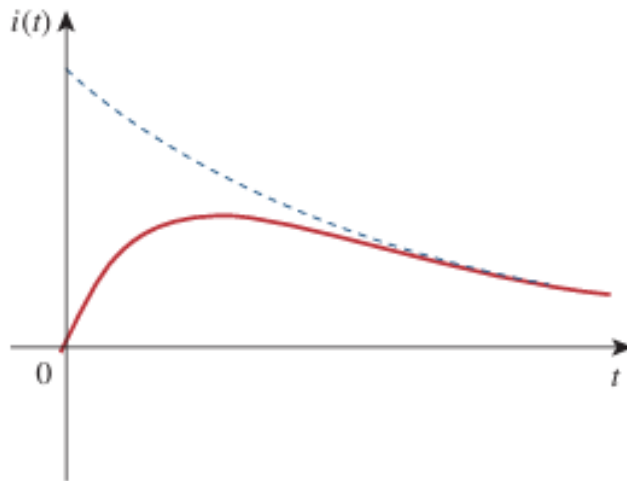
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Overdamped Case ($\alpha > \omega_0$):

Considering Eqn.8 and $\alpha > \omega_0$: $C > 4L/R^2$. When this happens, both roots are negative and real. The response is

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{--- 10}$$

The response decays and approaches zero as t increases.



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Critically Damped Case ($\alpha = \omega_0$):

Considering Eqn.8 and $\alpha = \omega_0$: $C = 4L/R^2$ and $s_1 = s_2 = -\alpha = -\frac{R}{2L}$

When this happens, the response will be like this,

$$i(t) = A_1 e^{-\alpha t} + A_2 e^{-\alpha t} = A_3 e^{-\alpha t}$$

Where $A_1 + A_2 = A_3$. This cannot be the solution, because the two initial conditions cannot be satisfied with the single constant.

This means that our **assumption of an exponential solution is incorrect for the special case of critical damping.**

When $\alpha = \omega_0 = R/2L$, Eqn.3 becomes

$$\frac{d^2 i}{dt^2} + 2\alpha \frac{di}{dt} + \alpha^2 i = 0$$

$$\frac{d}{dt} \left(\frac{di}{dt} + \alpha i \right) + \alpha \left(\frac{di}{dt} + \alpha i \right) = 0 \quad \text{--- 11}$$

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If we let $f = \frac{di}{dt} + \alpha i$ Eqn.11 becomes, $\frac{df}{dt} + \alpha f = 0$ This is a first order equation

with solution $f = A_1 e^{-\alpha t}$, where A_1 is a constant.

Now

$$\frac{di}{dt} + \alpha i = A_1 e^{-\alpha t}$$

$$e^{\alpha t} \frac{di}{dt} + e^{\alpha t} \alpha i = A_1$$

$$\frac{d}{dt}(e^{\alpha t} i) = A_1$$

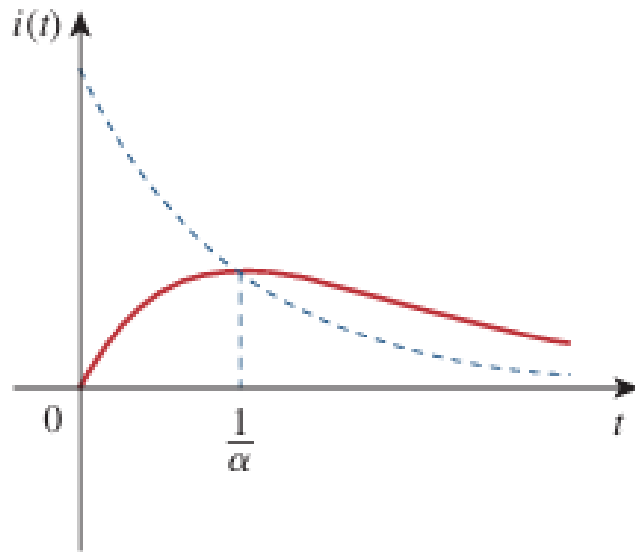
Integrating both sides yields, $e^{\alpha t} i = A_1 t + A_2$ where A_2 is another constant.

$$i = (A_1 t + A_2) e^{-\alpha t}$$

Hence, the natural response of the critically damped circuit is a **sum of two terms: a negative exponential and a negative exponential multiplied by a linear term.**

$$i(t) = (A_2 + A_1 t) e^{-\alpha t}$$

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$$i(t) = (A_2 + A_1 t)e^{-\alpha t}$$

The response curve is a sketch of $i(t) = t e^{-\alpha t}$, which reaches a maximum value of e^{-t}/α at $t = 1/\alpha$, one time constant, and then decays all the way to zero.

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Underdamped Case ($\alpha < \omega_0$):

Considering Eqn.8 and $\alpha < \omega_0$: $C < 4L/R^2$.

$$s_1 = -\alpha + \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha + j\omega_d$$

$$s_2 = -\alpha - \sqrt{-(\omega_0^2 - \alpha^2)} = -\alpha - j\omega_d$$

where $j = \sqrt{-1}$ and $\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$ which is the damping frequency. ω_0 is often called the undamped natural frequency, ω_d is called the damped natural frequency. The natural response is:

$$\begin{aligned} i(t) &= A_1 e^{-(\alpha - j\omega_d)t} + A_2 e^{-(\alpha + j\omega_d)t} \\ &= e^{-\alpha t} (A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t}) \end{aligned}$$

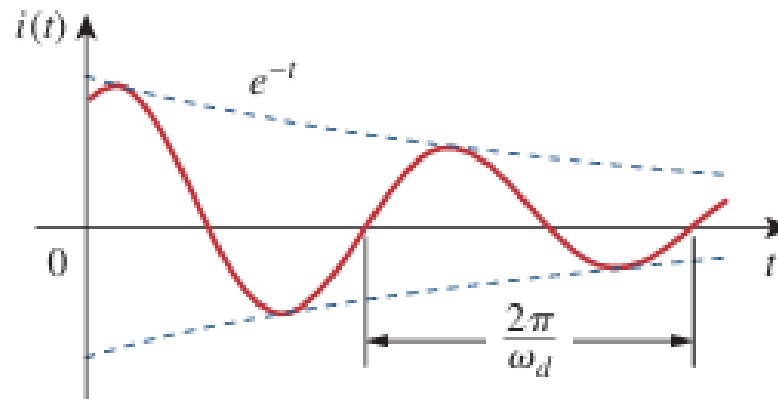
Using Euler's identity,

$$\begin{aligned} i(t) &= e^{-\alpha t} [A_1 (\cos \omega_d t + j \sin \omega_d t) + A_2 (\cos \omega_d t - j \sin \omega_d t)] \\ &= e^{-\alpha t} [(A_1 + A_2) \cos \omega_d t + j(A_1 - A_2) \sin \omega_d t] \end{aligned}$$

$$i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

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With the presence of sine and cosine functions, it is clear that the natural response for this case is exponentially damped and oscillatory in nature. The response has a time constant of $1/\alpha$ and a period of $T = 2\pi/\omega_d$



Once the inductor current is found for the RLC series circuit, other circuit quantities such as individual element voltages can easily be found.

$$v_R = Ri, \quad v_L = Ldi/dt$$

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Three important points to be observed:

1. The damping effect is due to the presence of resistance R . The damping factor α determines the rate at which the response is damped.

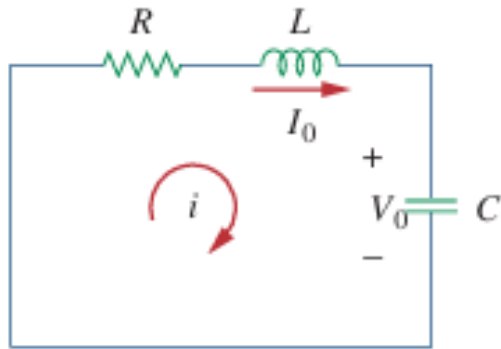
If $R = 0$, then $\alpha = 0$ and we have an LC circuit with $1/\sqrt{LC}$ as the undamped natural frequency. Since $\alpha < \omega_0$ in this case, the response is not only undamped but also oscillatory. The circuit is said to be *loss-less*, because the dissipating or damping element (R) is absent.

2. Oscillatory response is possible due to the presence of the two types of storage elements. Having both L and C allows the flow of energy back and forth between the two. The damped oscillation exhibited by the underdamped response is known as *ringing*.

3. The critically damped case is the borderline between the underdamped and overdamped cases and it decays the fastest. With the same initial conditions, the overdamped case has the longest settling time, because it takes the longest time to dissipate the initial stored energy.

Exercise

In the circuit, $R = 40\ \Omega$, $L = 4\text{H}$ and $C = \frac{1}{4}\text{F}$, calculate the characteristic roots of the circuit. Is the natural response overdamped, under damped, or critically damped?



$$\alpha = \frac{R}{2L} = \frac{40}{2(4)} = 5, \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \times \frac{1}{4}}} = 1$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -5 \pm \sqrt{25 - 1}$$

$$s_1 = -0.101, \quad s_2 = -9.899$$

Since $\alpha > \omega_0$ we conclude that the response is overdamped. This is also evident from the fact that the roots are real and negative.

Next class: Source Free RLC parallel circuit